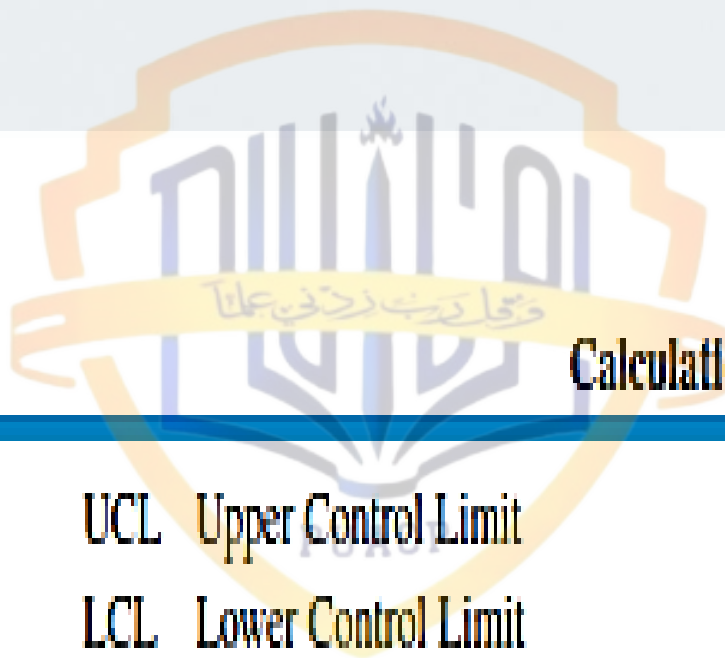




Control Charts for Variables



SPC (Statistical Process Control)

Calculations for Control Limits

Notation:	UCL	Upper Control Limit	$\bar{x}$	Average of Measurements
	LCL	Lower Control Limit	$\bar{\bar{x}}$	Average of Averages
	CL	Center Line	R	Range
	n	Sample Size	$\bar{R}$	Average of Ranges
	PCR	Process Capability Ratio	USL	Upper Specification Limit
	$\hat{\sigma}$	Process Standard Deviation	LSL	Lower Specification Limit

Variables Data ($\bar{X}$ and R Control Charts)

$\bar{X}$ Control Chart

$$UCL = \bar{\bar{x}} + A_2 \bar{R}$$

$$LCL = \bar{\bar{x}} - A_2 \bar{R}$$

$$CL = \bar{\bar{x}}$$

R Control Chart

$$UCL = \bar{R} D_4$$

$$LCL = \bar{R} D_3$$

$$CL = \bar{R}$$

Capability Study

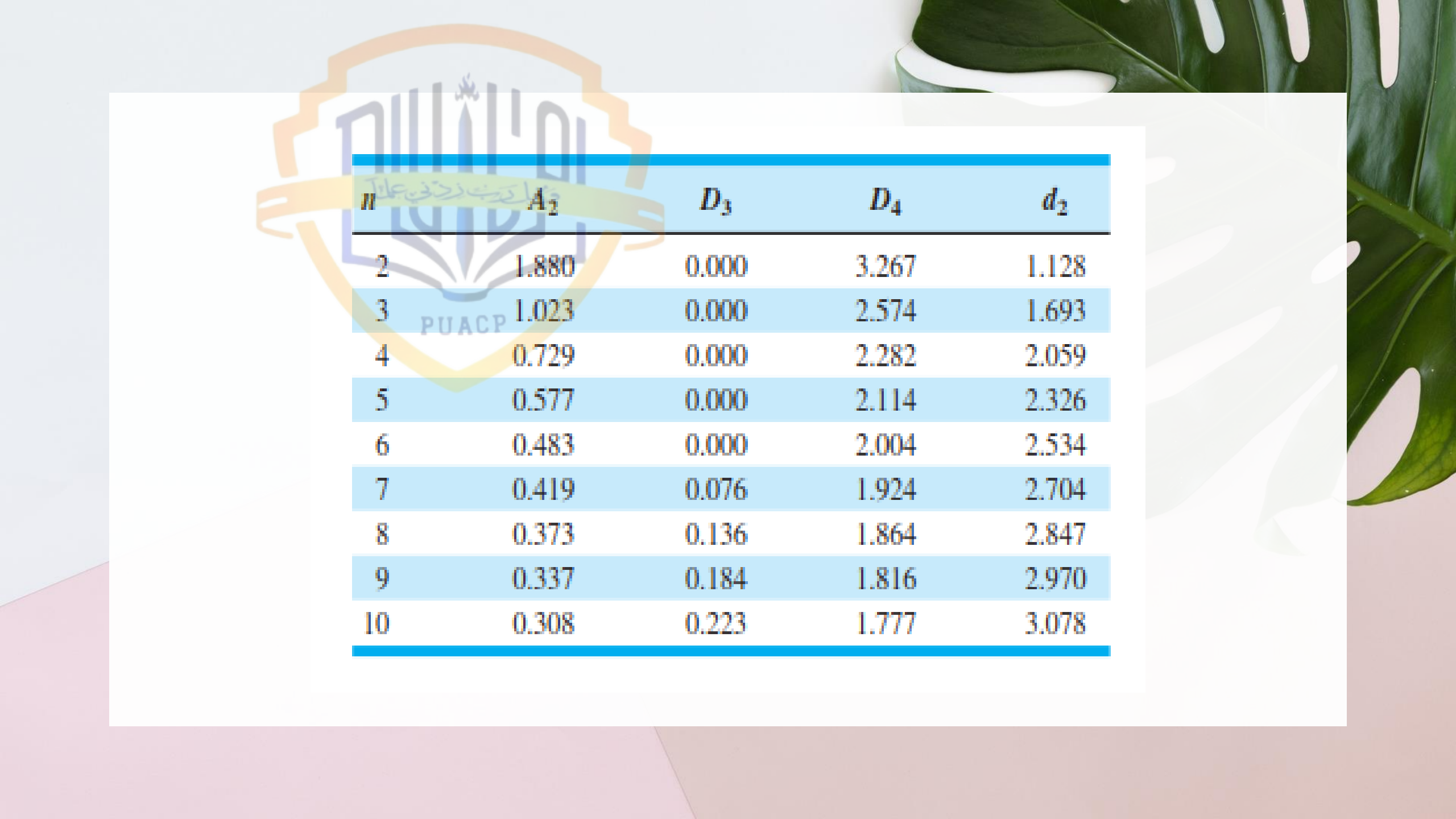
$$C_p = (USL - LSL)/(6 \hat{\sigma}); \text{ where } \hat{\sigma} = \bar{R}/d_2$$

Attribute Data (p , np , c , and u Control Charts)

Control Chart Formulas

	p (fraction)	np (number of nonconforming)	c (count of nonconformances)	u (count of nonconformances/unit)
CL	$\bar{p}$	$n\bar{p}$	$\bar{c}$	$\bar{u}$
UCL	$\bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$	$n\bar{p} + 3\sqrt{n\bar{p}(1-\bar{p})}$	$\bar{c} + 3\sqrt{\bar{c}}$	$\bar{u} + 3\sqrt{\frac{\bar{u}}{n}}$
LCL	$\bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$	$n\bar{p} - 3\sqrt{n\bar{p}(1-\bar{p})}$	$\bar{c} - 3\sqrt{\bar{c}}$	$\bar{u} - 3\sqrt{\frac{\bar{u}}{n}}$
Notes	If n varies, use π or individual n_i	n must be a constant	n must be a constant	If n varies, use π or individual n_i

→ $\bar{n}$



n	A_2	D_3	D_4	d_2
2	1.880	0.000	3.267	1.128
3	1.023	0.000	2.574	1.693
4	0.729	0.000	2.282	2.059
5	0.577	0.000	2.114	2.326
6	0.483	0.000	2.004	2.534
7	0.419	0.076	1.924	2.704
8	0.373	0.136	1.864	2.847
9	0.337	0.184	1.816	2.970
10	0.308	0.223	1.777	3.078

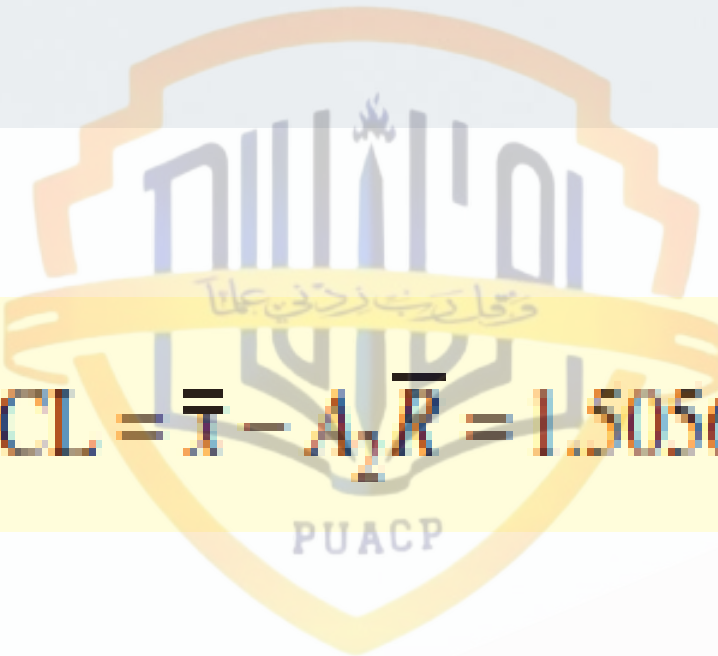
Flow Width Measurements (microns) for the Hard-Bake Process

Sample Number	Wafers					$\bar{x}_i$	R_i
	1	2	3	4	5		
1	1.3235	1.4128	1.6744	1.4573	1.6914	1.5119	0.3679
2	1.4314	1.3592	1.6075	1.4666	1.6109	1.4951	0.2517
3	1.4284	1.4871	1.4932	1.4324	1.5674	1.4817	0.1390
4	1.5028	1.6352	1.3841	1.2831	1.5507	1.4712	0.3521
5	1.5604	1.2735	1.5265	1.4363	1.6441	1.4882	0.3706
6	1.5955	1.5451	1.3574	1.3281	1.4198	1.4492	0.2674
7	1.6274	1.5064	1.8366	1.4177	1.5144	1.5805	0.4189
8	1.4190	1.4303	1.6637	1.6067	1.5519	1.5343	0.2447
9	1.3884	1.7277	1.5355	1.5176	1.3688	1.5076	0.3589
10	1.4039	1.6697	1.5089	1.4627	1.5220	1.5134	0.2658
11	1.4158	1.7667	1.4278	1.5928	1.4181	1.5242	0.3509
12	1.5821	1.3355	1.5777	1.3908	1.7559	1.5284	0.4204
13	1.2856	1.4106	1.4447	1.6398	1.1928	1.3947	0.4470
14	1.4951	1.4036	1.5893	1.6458	1.4969	1.5261	0.2422
15	1.3589	1.2863	1.5996	1.2497	1.5471	1.4083	0.3499
16	1.5747	1.5301	1.5171	1.1839	1.8662	1.5344	0.6823
17	1.3680	1.7269	1.3957	1.5014	1.4449	1.4874	0.3589
18	1.4163	1.3864	1.3057	1.6210	1.5573	1.4573	0.3153
19	1.5796	1.4185	1.6541	1.5116	1.7247	1.5777	0.3062
20	1.7106	1.4412	1.2361	1.3820	1.7601	1.5060	0.5240
21	1.4371	1.5051	1.3485	1.5670	1.4880	1.4691	0.2185
22	1.4738	1.5936	1.6583	1.4973	1.4720	1.5390	0.1863
23	1.5917	1.4333	1.5551	1.5295	1.6866	1.5592	0.2533
24	1.6399	1.5243	1.5705	1.5563	1.5530	1.5688	0.1156
25	1.5797	1.3663	1.6240	1.3732	1.6887	1.5264	0.3224
$\Sigma \bar{x}_i = 37.6400$						$\Sigma R_i = 8.1302$	
$\bar{\bar{x}} = 1.5056$						$\bar{R} = 0.32521$	

Development and Use of $\bar{x}$ and R Charts

$$\bar{\bar{x}} = \frac{\sum_{i=1}^{25} \bar{x}_i}{25} = \frac{37.6400}{25} = 1.5056$$

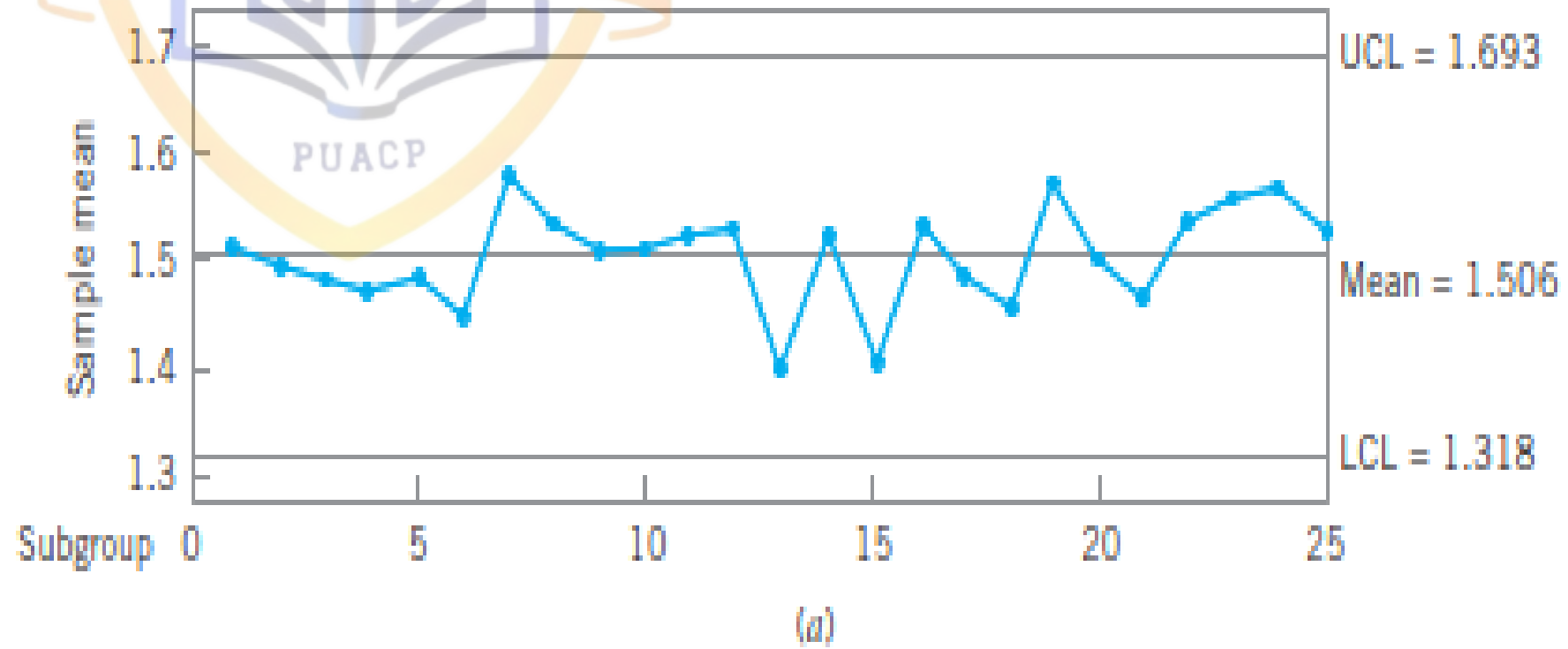
$$\bar{R} = \frac{\sum_{i=1}^{25} R_i}{25} = \frac{8.1302}{25} = 0.32521$$

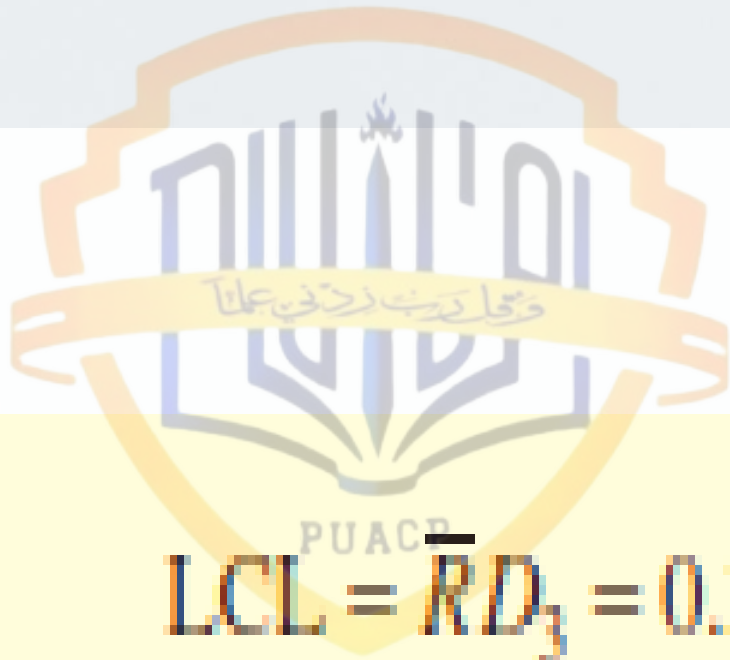


The logo of the University of Poonchri, Al-Qadisiyah (PUACP) is a shield-shaped emblem. It features a central minaret with a flame on top, flanked by two stylized trees. The Arabic text 'جامعة بونشري القادسية' is written in a banner across the middle, and 'PUACP' is written at the bottom.

$$LCL = \bar{\bar{x}} - A_2 \bar{R} = 1.5056 - (0.577)(0.32521) = 1.31795$$

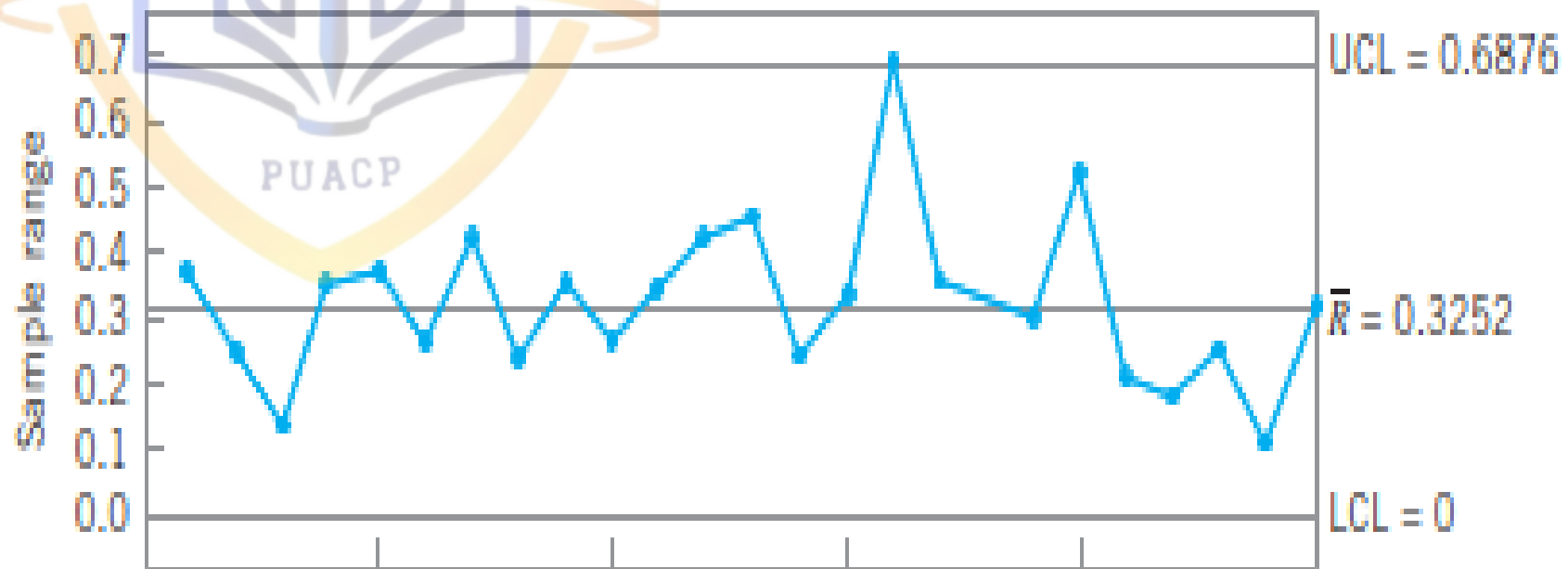
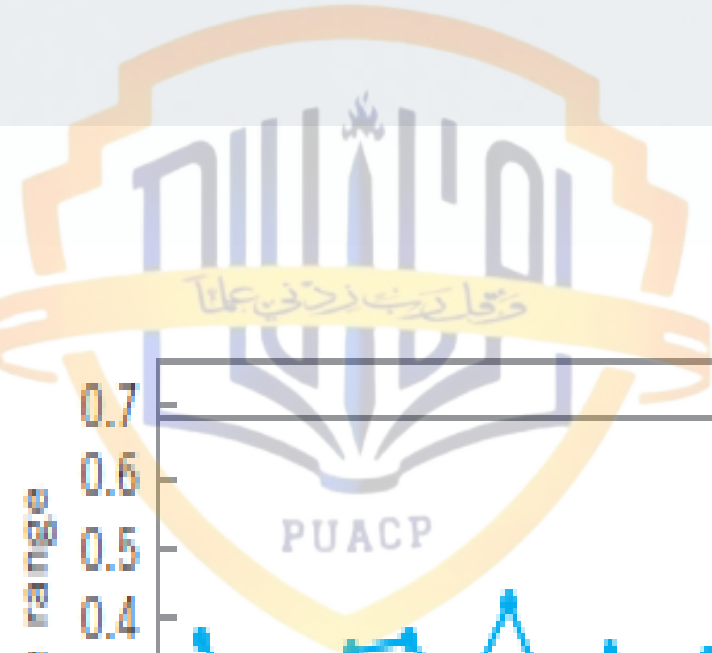
$$UCL = \bar{\bar{x}} + A_2 \bar{R} = 1.5056 + (0.577)(0.32521) = 1.69325$$





$$LCL = \bar{RD}_3 = 0.32521(0) = 0$$

$$UCL = \bar{RD}_4 = 0.32521(2.114) = 0.68749$$



(h)

Estimating Process Capability.

$$\hat{\sigma} = \frac{\bar{R}}{d_2} = \frac{0.32521}{2.326} = 0.1398 \text{ microns}$$

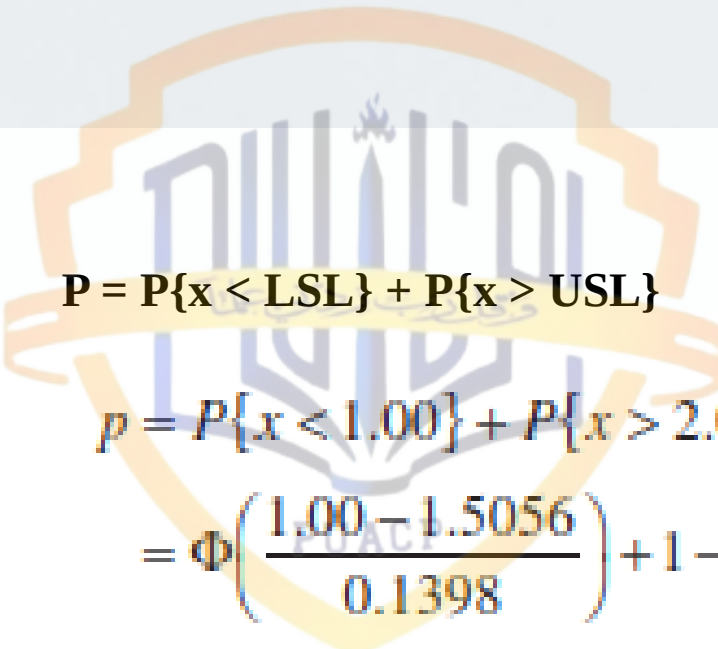
$d_2 = 2.326$ with sample size 5
Using table vi page 720

$$\bar{\bar{x}} \pm 3\hat{\sigma}$$

$$1.50 \pm 0.50$$

with mean 1.5056 and standard deviation 0.1398,

$$LSL = 1.5 - 0.5 = 1, \quad USL = 1.5 + 0.5 = 2$$


$$P = P\{x < LSL\} + P\{x > USL\}$$

$$\begin{aligned} p &= P\{x < 1.00\} + P\{x > 2.00\} \\ &= \Phi\left(\frac{1.00 - 1.5056}{0.1398}\right) + 1 - \Phi\left(\frac{2.00 - 1.5056}{0.1398}\right) \\ &= \Phi(-3.61660) + 1 - \Phi(3.53648) \\ &\approx 0.00015 + 1 - 0.99980 \\ &\approx 0.00035 \end{aligned}$$

That is, about 0.035% [350 parts per million (ppm)] of the wafers produced will be outside of the specifications.

process capability ratio (PCR) C_p

Another way to express process capability is in terms of the **process capability ratio**

$$C_p = \frac{USL - LSL}{6\sigma}$$

$$\hat{C}_p = \frac{2.00 - 1.00}{6(0.1398)} = \frac{1.00}{0.8388} = 1.192$$

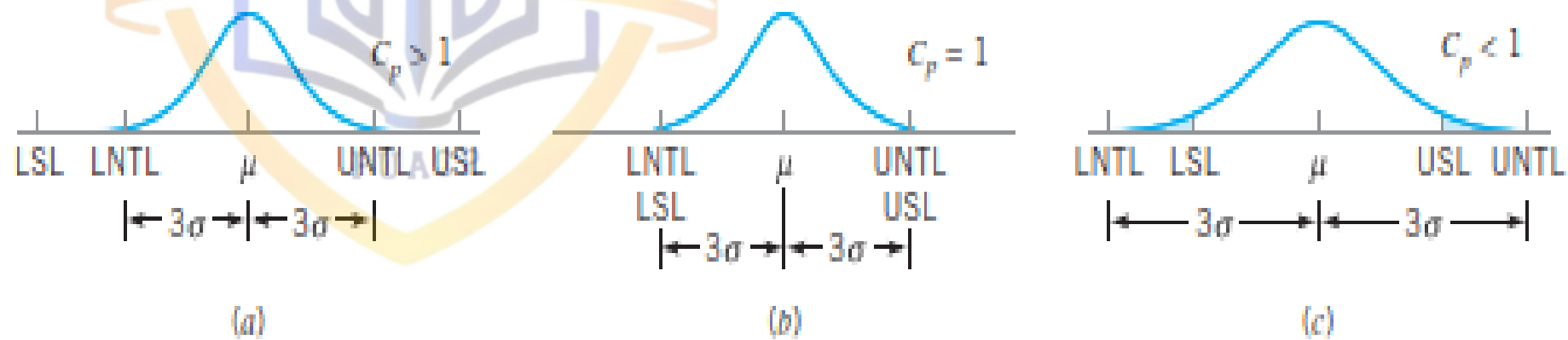
This implies that the “natural” tolerance limits in the process (three-sigma above and below the mean) are inside the lower and upper specification limits. Consequently, a moderately small number of nonconforming wafers will be produced. The PCR C_p may be interpreted another way. The quantity



$$P = \left(\frac{1}{C_p} \right) 100\%$$

$$\hat{P} = \left(\frac{1}{\hat{C}_p} \right) 100\% = \left(\frac{1}{1.192} \right) 100\% = 83.89$$

That is, the process uses up about 84% of the specification band.



cations. In Figure 6.3a the PCR C_p is greater than unity. This means that the process uses up much less than 100% of the tolerance band. Consequently, relatively few nonconforming units

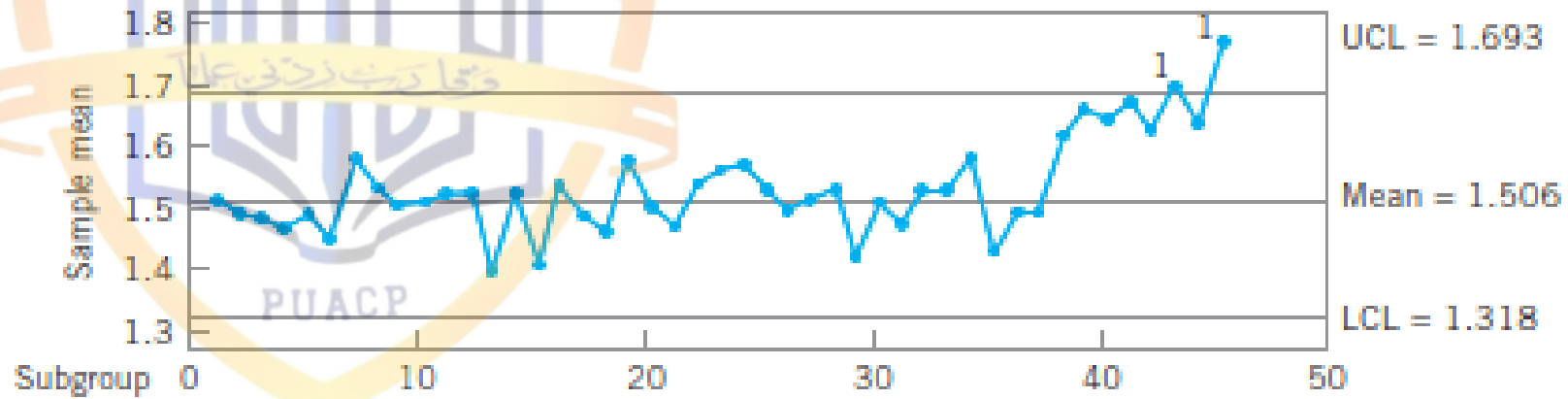


will be produced by this process. Figure 6.3*b* shows a process for which the PCR $C_p = 1$; that is, the process uses up all the tolerance band. For a normal distribution this would imply about 0.27% (or 2,700 ppm) nonconforming units. Finally, Figure 6.3*c* presents a process for which the PCR $C_p < 1$; that is, the process uses up more than 100% of the tolerance band. In this case, the process is very yield-sensitive, and a large number of nonconforming units will be produced.

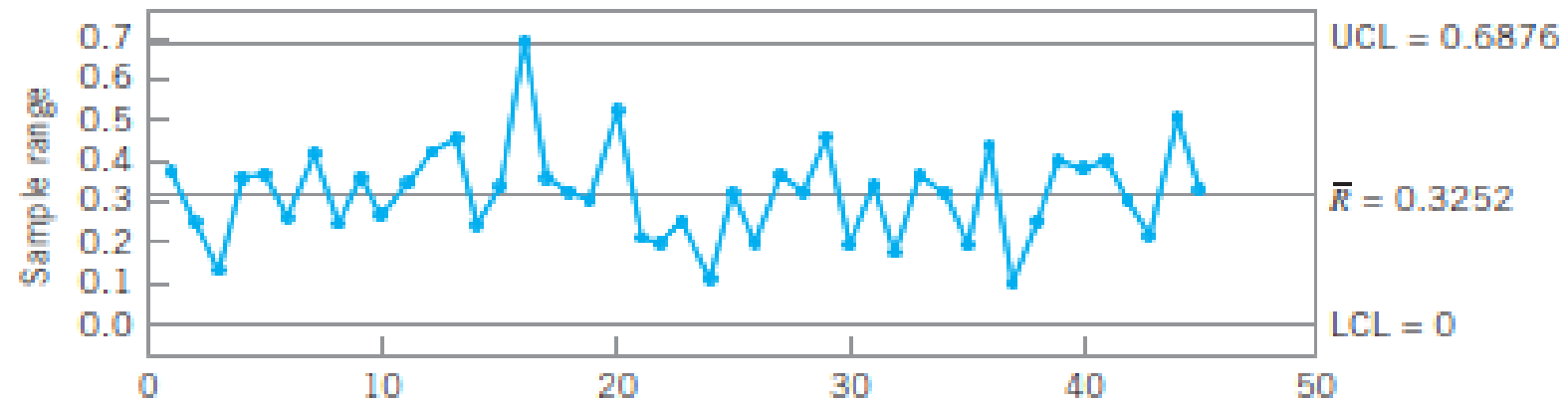
■ **TABLE 6.2**

Additional Samples for Example 6.1

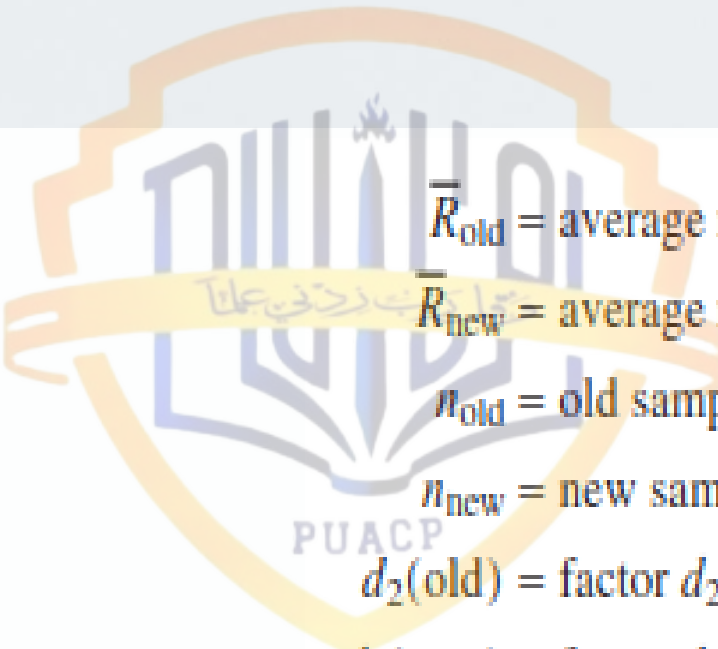
Sample Number	Wafers					$\bar{x}_i$	R_i
	1	2	3	4	5		
26	1.4483	1.5458	1.4538	1.4303	1.6206	1.4998	0.1903
27	1.5435	1.6899	1.5830	1.3358	1.4187	1.5142	0.3541
28	1.5175	1.3446	1.4723	1.6657	1.6661	1.5332	0.3215
29	1.5454	1.0931	1.4072	1.5039	1.5264	1.4152	0.4523
30	1.4418	1.5059	1.5124	1.4620	1.6263	1.5097	0.1845
31	1.4301	1.2725	1.5945	1.5397	1.5252	1.4724	0.3220
32	1.4981	1.4506	1.6174	1.5837	1.4962	1.5292	0.1668
33	1.3009	1.5060	1.6231	1.5831	1.6454	1.5317	0.3445
34	1.4132	1.4603	1.5808	1.7111	1.7313	1.5793	0.3181
35	1.3817	1.3135	1.4953	1.4894	1.4596	1.4279	0.1818
36	1.5765	1.7014	1.4026	1.2773	1.4541	1.4824	0.4241
37	1.4936	1.4373	1.5139	1.4808	1.5293	1.4910	0.0920
38	1.5729	1.6738	1.5048	1.5651	1.7473	1.6128	0.2425
39	1.8089	1.5513	1.8250	1.4389	1.6558	1.6560	0.3861
40	1.6236	1.5393	1.6738	1.8698	1.5036	1.6420	0.3662
41	1.4120	1.7931	1.7345	1.6391	1.7791	1.6716	0.3811
42	1.7372	1.5663	1.4910	1.7809	1.5504	1.6252	0.2899
43	1.5971	1.7394	1.6832	1.6677	1.7974	1.6970	0.2003
44	1.4295	1.6536	1.9134	1.7272	1.4370	1.6321	0.4839
45	1.6217	1.8220	1.7915	1.6744	1.9404	1.7700	0.3187



(a)



(b)



$\bar{R}_{old}$ = average range for the old sample size

$\bar{R}_{new}$ = average range for the new sample size

n_{old} = old sample size

n_{new} = new sample size

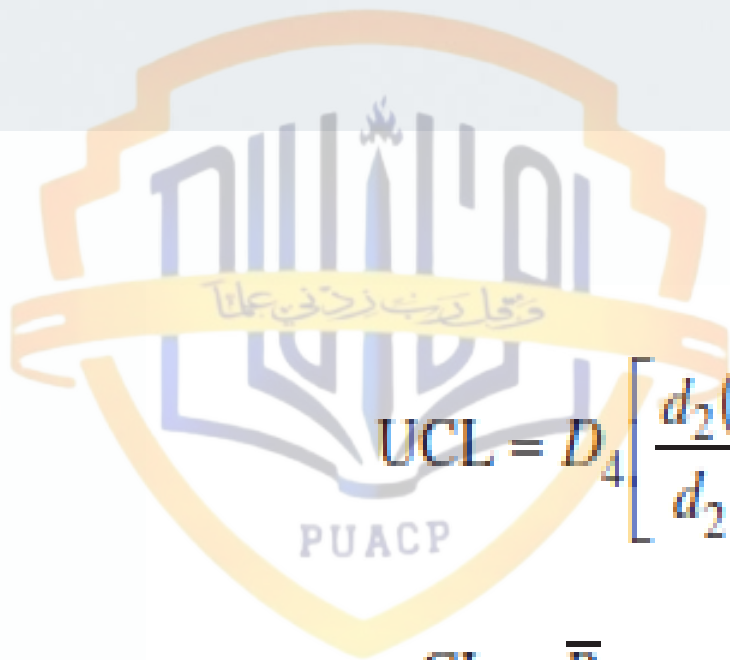
$d_2(old)$ = factor d_2 for the old sample size

$d_2(new)$ = factor d_2 for the new sample size

$$UCL = \bar{\bar{x}} + A_2 \left[\frac{d_2(new)}{d_2(old)} \right] \bar{R}_{old}$$

$$LCL = \bar{\bar{x}} - A_2 \left[\frac{d_2(new)}{d_2(old)} \right] \bar{R}_{old}$$

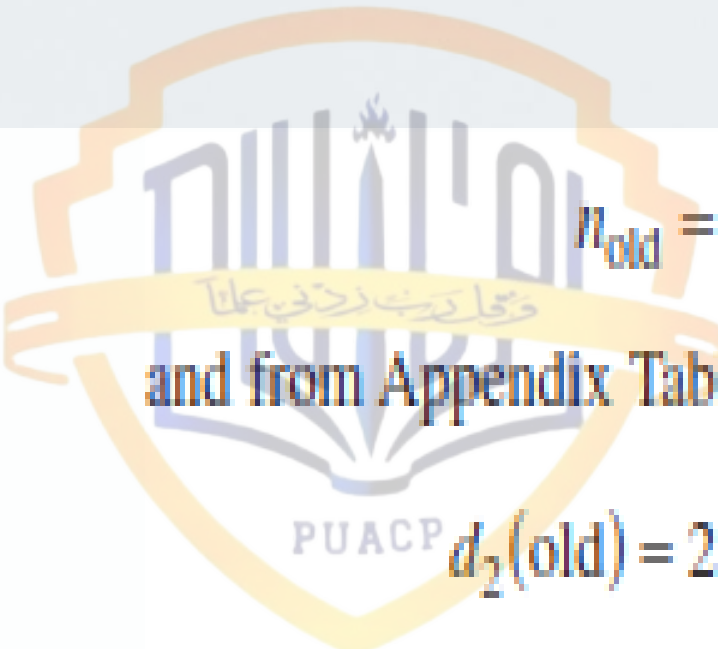
where the center line $\bar{\bar{x}}$ is unchanged and the factor A_2 is selected for the *new* sample size. For the R chart, the new parameters are



$$UCL = D_4 \left[\frac{d_2(\text{new})}{d_2(\text{old})} \right] \bar{R}_{\text{old}}$$

$$CL = \bar{R}_{\text{new}} = \left[\frac{d_2(\text{new})}{d_2(\text{old})} \right] \bar{R}_{\text{old}}$$

$$LCL = \max \left\{ 0, D_3 \left[\frac{d_2(\text{new})}{d_2(\text{old})} \right] \bar{R}_{\text{old}} \right\}$$

The logo of the Prince Sultan bin Abdul Aziz University for Advanced College of Pharmacy (PUACP) is located in the top left corner. It features a shield-like emblem with Arabic calligraphy and the acronym 'PUACP' at the bottom.
$$n_{\text{old}} = 5 \quad \bar{R}_{\text{old}} = 0.32521$$

and from Appendix Table VI we have

$$d_2(\text{old}) = 2.326 \quad d_2(\text{new}) = 1.693$$

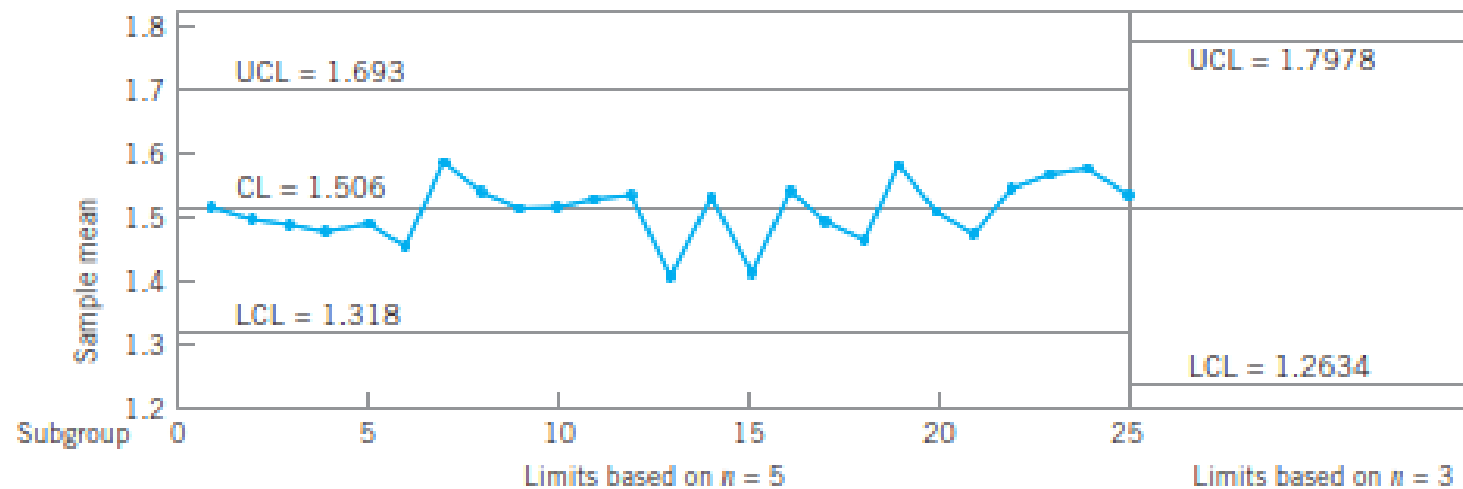
Therefore, the new control limits on the $\bar{x}$ chart are found

$$\begin{aligned} \text{UCL} &= \bar{\bar{x}} + A_2 \left[\frac{d_2(\text{new})}{d_2(\text{old})} \right] \bar{R}_{\text{old}} \\ &= 1.5056 + (1.023) \left[\frac{1.693}{2.326} \right] (0.32521) \\ &= 1.5056 + 0.2422 = 1.7478 \end{aligned}$$

$$LCL = \bar{x} - A_2 \left[\frac{d_2(\text{new})}{d_2(\text{old})} \right] \bar{R}_{\text{old}}$$

$$= 1.5056 - (1.023) \left[\frac{1.693}{2.326} \right] (0.32521)$$

$$= 1.5056 - 0.2422 = 1.2634$$

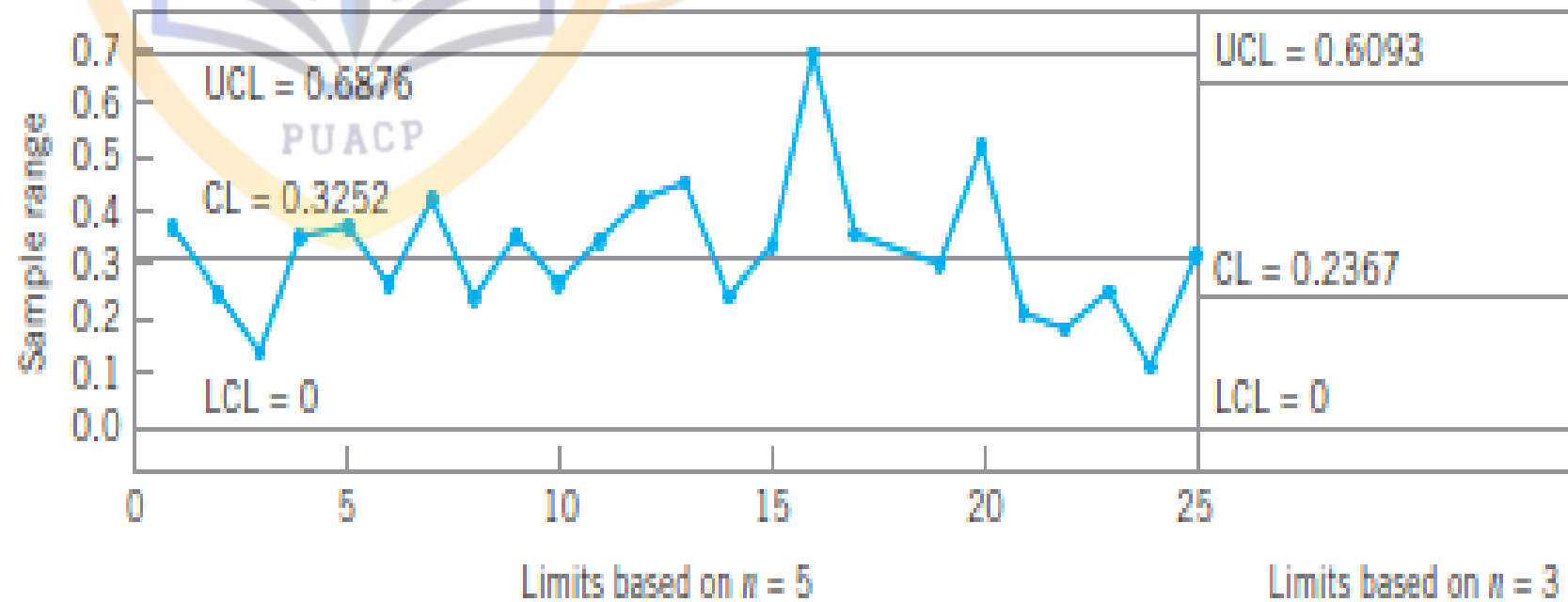


For the R chart, the new parameters are given by equation

$$\begin{aligned} \text{UCL} &= D_4 \left[\frac{d_2(\text{new})}{d_2(\text{old})} \right] \bar{R}_{\text{old}} \\ &= (2.574) \left[\frac{1.693}{2.326} \right] (0.32521) \\ &= 0.6093 \end{aligned}$$

$$\begin{aligned} \text{CL} &= \bar{R}_{\text{new}} = \left[\frac{d_2(\text{new})}{d_2(\text{old})} \right] \bar{R}_{\text{old}} \\ &= \left[\frac{1.693}{2.326} \right] (0.32521) \\ &= 0.2367 \end{aligned}$$

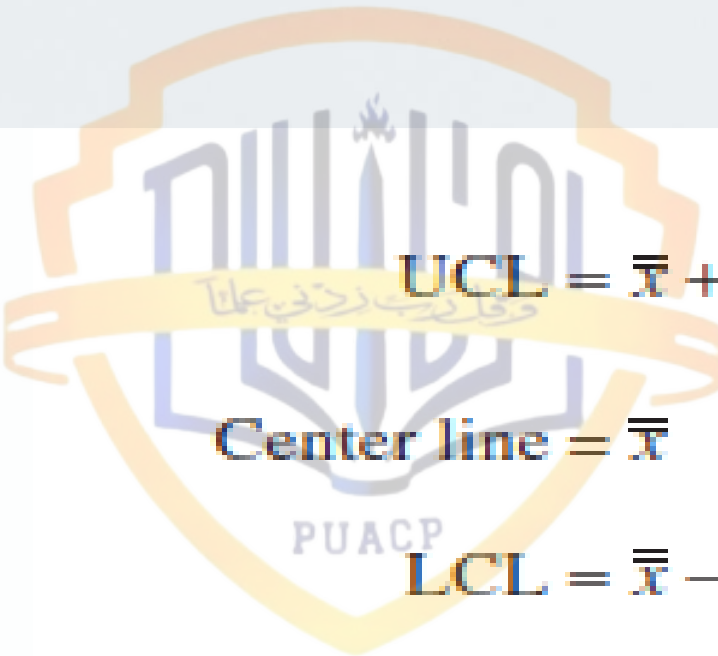
$$\begin{aligned} \text{LCL} &= \max \left\{ 0, D_3 \left[\frac{d_2(\text{new})}{d_2(\text{old})} \right] \bar{R}_{\text{old}} \right\} \\ &= 0 \end{aligned}$$



Inside Diameter Measurements (mm) for Automobile Engine Piston Rings

Sample Number	Observations					$\bar{x}_i$	s_i
1	74.030	74.002	74.019	73.992	74.008	74.010	0.0148
2	73.995	73.992	74.001	74.011	74.004	74.001	0.0075
3	73.988	74.024	74.021	74.005	74.002	74.008	0.0147
4	74.002	73.996	73.993	74.015	74.009	74.003	0.0091
5	73.992	74.007	74.015	73.989	74.014	74.003	0.0122
6	74.009	73.994	73.997	73.985	73.993	73.996	0.0087
7	73.995	74.006	73.994	74.000	74.005	74.000	0.0055
8	73.985	74.003	73.993	74.015	73.988	73.997	0.0123
9	74.008	73.995	74.009	74.005	74.004	74.004	0.0055
10	73.998	74.000	73.990	74.007	73.995	73.998	0.0063
11	73.994	73.998	73.994	73.995	73.990	73.994	0.0029
12	74.004	74.000	74.007	74.000	73.996	74.001	0.0042
13	73.983	74.002	73.998	73.997	74.012	73.998	0.0105
14	74.006	73.967	73.994	74.000	73.984	73.990	0.0153
15	74.012	74.014	73.998	73.999	74.007	74.006	0.0073
16	74.000	73.984	74.005	73.998	73.996	73.997	0.0078
17	73.994	74.012	73.986	74.005	74.007	74.001	0.0106
18	74.006	74.010	74.018	74.003	74.000	74.007	0.0070
19	73.984	74.002	74.003	74.005	73.997	73.998	0.0085
20	74.000	74.010	74.013	74.020	74.003	74.009	0.0080
21	73.982	74.001	74.015	74.005	73.996	74.000	0.0122
22	74.004	73.999	73.990	74.006	74.009	74.002	0.0074
23	74.010	73.989	73.990	74.009	74.014	74.002	0.0119
24	74.015	74.008	73.993	74.000	74.010	74.005	0.0087
25	73.982	73.984	73.995	74.017	74.013	73.998	0.0162
						$\Sigma = 1,850.028$	0.2351
						$\bar{\bar{x}} = 74.001$	$\bar{s} = 0.0094$

For Mean



The logo of the Punjab University of Agriculture, Chokkistan (PUACP) is located in the top left corner. It features a shield-shaped emblem with a yellow border. Inside the shield, there is a blue and white geometric design. A yellow banner with Arabic calligraphy is draped across the middle of the shield. Below the shield, the acronym 'PUACP' is written in blue capital letters.

$$\text{UCL} = \bar{\bar{x}} + \frac{3\bar{s}}{c_4\sqrt{n}}$$

$$\text{Center line} = \bar{\bar{x}}$$

$$\text{LCL} = \bar{\bar{x}} - \frac{3\bar{s}}{c_4\sqrt{n}}$$

$$\text{UCL} = \bar{\bar{x}} + A_3\bar{s}$$

$$\text{Center line} = \bar{\bar{x}}$$

$$\text{LCL} = \bar{\bar{x}} - A_3\bar{s}$$

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$$

For S. D

$$UCL = B_4 \bar{s}$$

$$\text{Center line} = \bar{x}$$

$$LCL = B_3 \bar{s}$$

$$\bar{\bar{x}} = \frac{1}{25} \sum_{i=1}^{25} \bar{x}_i = \frac{1}{25} (1,850.028) = 74.001$$

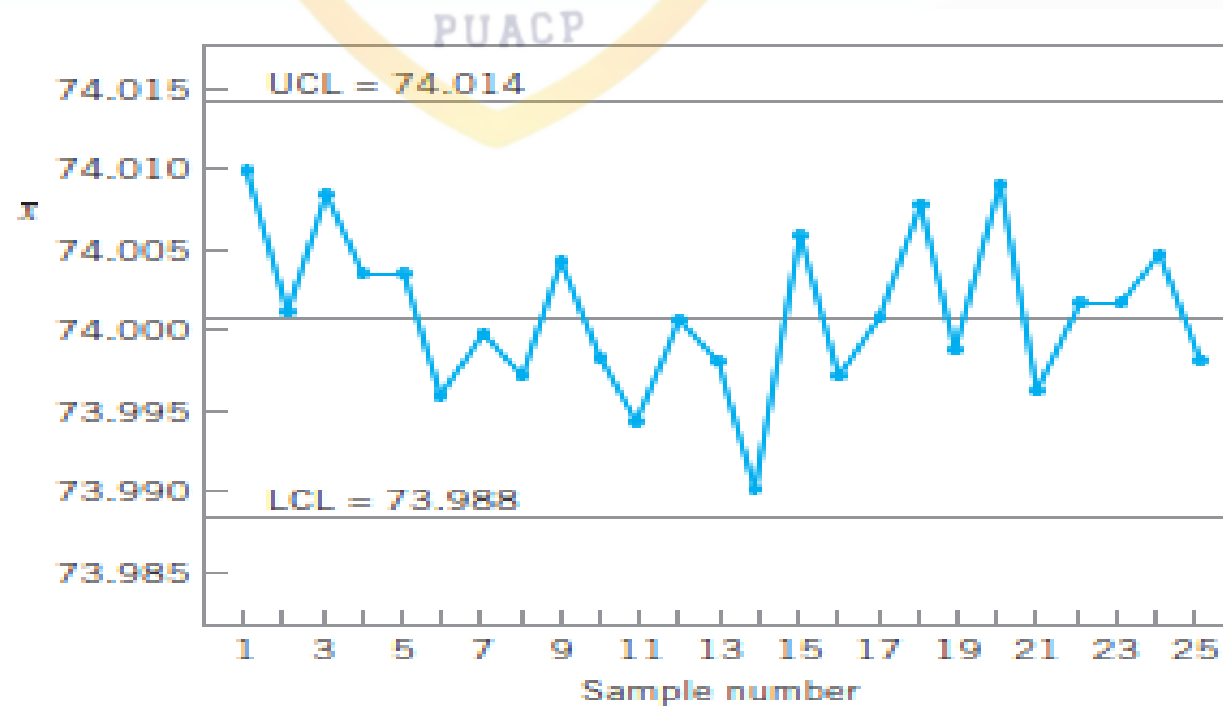
and

$$\bar{s} = \frac{1}{25} \sum_{i=1}^{25} s_i = \frac{1}{25} (0.2351) = 0.0094$$

$$UCL = \bar{\bar{x}} + A_3\bar{s} = 74.001 + (1.427)(0.0094) = 74.014$$

$$CL = \bar{\bar{x}} = 74.001$$

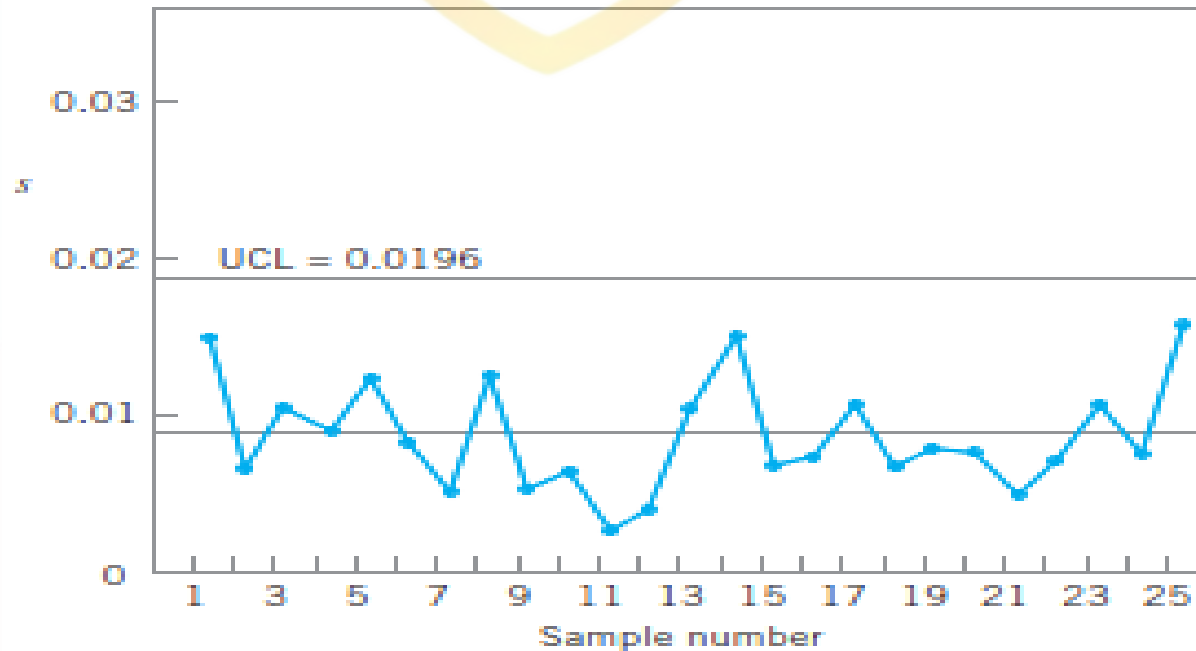
$$LCL = \bar{\bar{x}} - A_3\bar{s} = 74.001 - (1.427)(0.0094) = 73.988$$



$$UCL = B_4 \bar{s} = (2.089)(0.0094) = 0.0196$$

$$CL = \bar{s} = 0.0094$$

$$LCL = B_3 \bar{s} = (0)(0.0094) = 0$$



The $\bar{x}$ and s Control Charts with Variable Sample Size

Inside Diameter Measurements (mm) on Automobile Engine Piston Rings

Sample Number	Observations					$\bar{x}_i$	s_i
1	74.030	74.002	74.019	73.992	74.008	74.010	0.0148
2	73.995	73.992	74.001			73.996	0.0046
3	73.988	74.024	74.021	74.005	74.002	74.008	0.0147
4	74.002	73.996	73.993	74.015	74.009	74.003	0.0091
5	73.992	74.007	74.015	73.989	74.014	74.003	0.0122
6	74.009	73.994	73.997	73.985		73.996	0.0099
7	73.995	74.006	73.994	74.000		73.999	0.0055
8	73.985	74.003	73.993	74.015	73.988	73.997	0.0123
9	74.008	73.995	74.009	74.005		74.004	0.0064
10	73.998	74.000	73.990	74.007	73.995	73.998	0.0063
11	73.994	73.998	73.994	73.995	73.990	73.994	0.0029
12	74.004	74.000	74.007	74.000	73.996	74.001	0.0042
13	73.983	74.002	73.998			73.994	0.0100
14	74.006	73.967	73.994	74.000	73.984	73.990	0.0153
15	74.012	74.014	73.998			74.008	0.0087
16	74.000	73.984	74.005	73.998	73.996	73.997	0.0078
17	73.994	74.012	73.986	74.005		73.999	0.0115
18	74.006	74.010	74.018	74.003	74.000	74.007	0.0070
19	73.984	74.002	74.003	74.005	73.997	73.998	0.0085
20	74.000	74.010	74.013			74.008	0.0068
21	73.982	74.001	74.015	74.005	73.996	74.000	0.0122
22	74.004	73.999	73.990	74.006	74.009	74.002	0.0074
23	74.010	73.989	73.990	74.009	74.014	74.002	0.0119
24	74.015	74.008	73.993	74.000	74.010	74.005	0.0087
25	73.982	73.984	73.995	74.017	74.013	73.998	0.0162

$$\begin{aligned}\bar{x} &= \frac{\sum_{l=1}^{25} n_l \bar{x}_l}{\sum_{l=1}^{25} n_l} = \frac{5(74.010) + 3(73.996) + \dots + 5(73.998)}{5 + 3 + \dots + 5} \\ &= \frac{8,362.075}{113} = 74.001\end{aligned}$$

$$\begin{aligned}s &= \left[\frac{\sum_{l=1}^{25} (n_l - 1) s_l^2}{\sum_{l=1}^{25} n_l - 25} \right]^{1/2} = \left[\frac{4(0.0148)^2 + 2(0.0046)^2 + \dots + 4(0.0162)^2}{5 + 3 + \dots + 5 - 25} \right]^{1/2} \\ &= \left[\frac{0.009324}{88} \right]^{1/2} = 0.0103\end{aligned}$$

For Mean

$$UCL = \bar{\bar{x}} + \frac{3\bar{s}}{c_4\sqrt{n}}$$

$$\text{Center line} = \bar{\bar{x}}$$

$$LCL = \bar{\bar{x}} - \frac{3\bar{s}}{c_4\sqrt{n}}$$

$$UCL = 74.001 + (1.427)(0.0103) = 74.016$$

$$CL = 74.001$$

$$LCL = 74.001 - (1.427)(0.0103) = 73.986$$

For S. D

$$UCL = B_4 \bar{s}$$

$$\text{Center line} = \bar{s}$$

$$LCL = B_3 \bar{s}$$

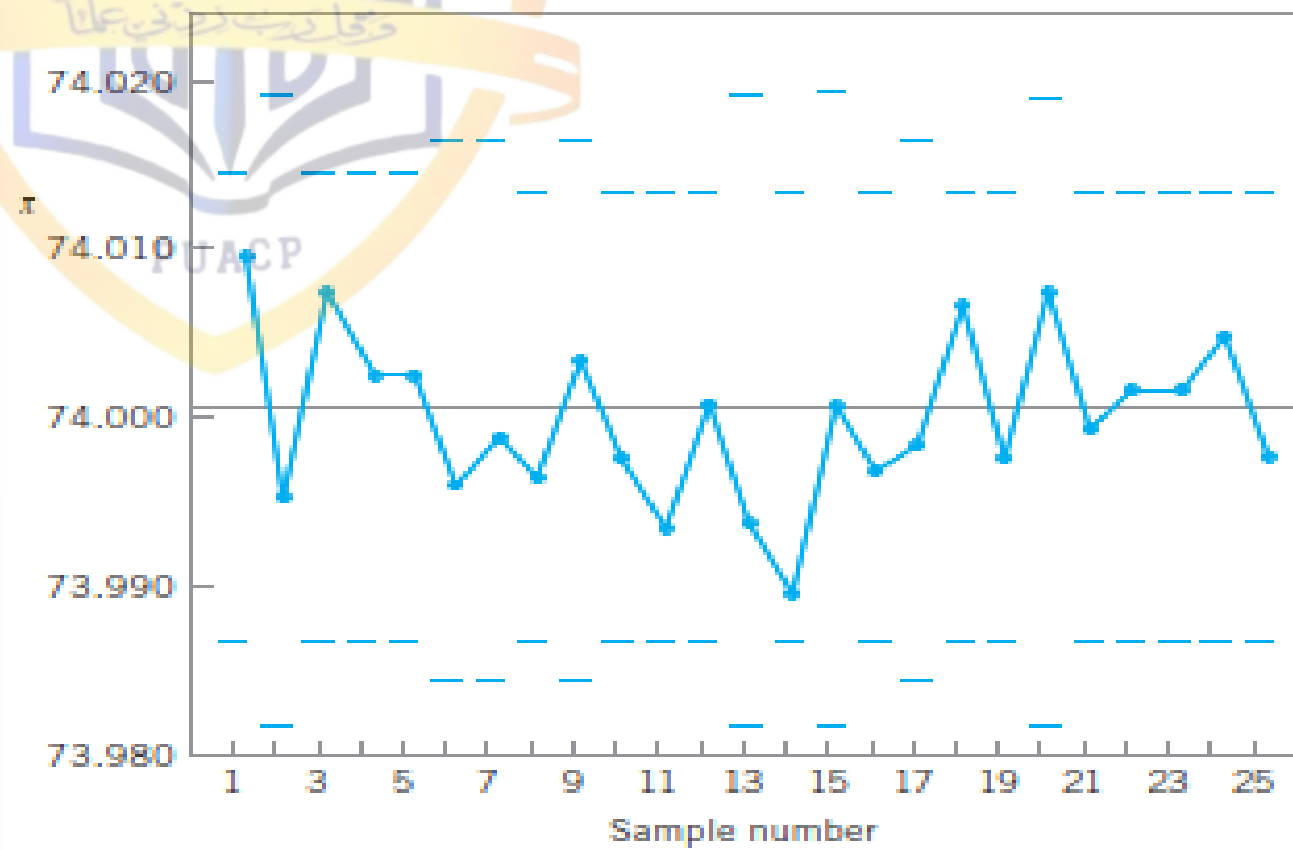
$$UCL = (2.089)(0.0103) = 0.022$$

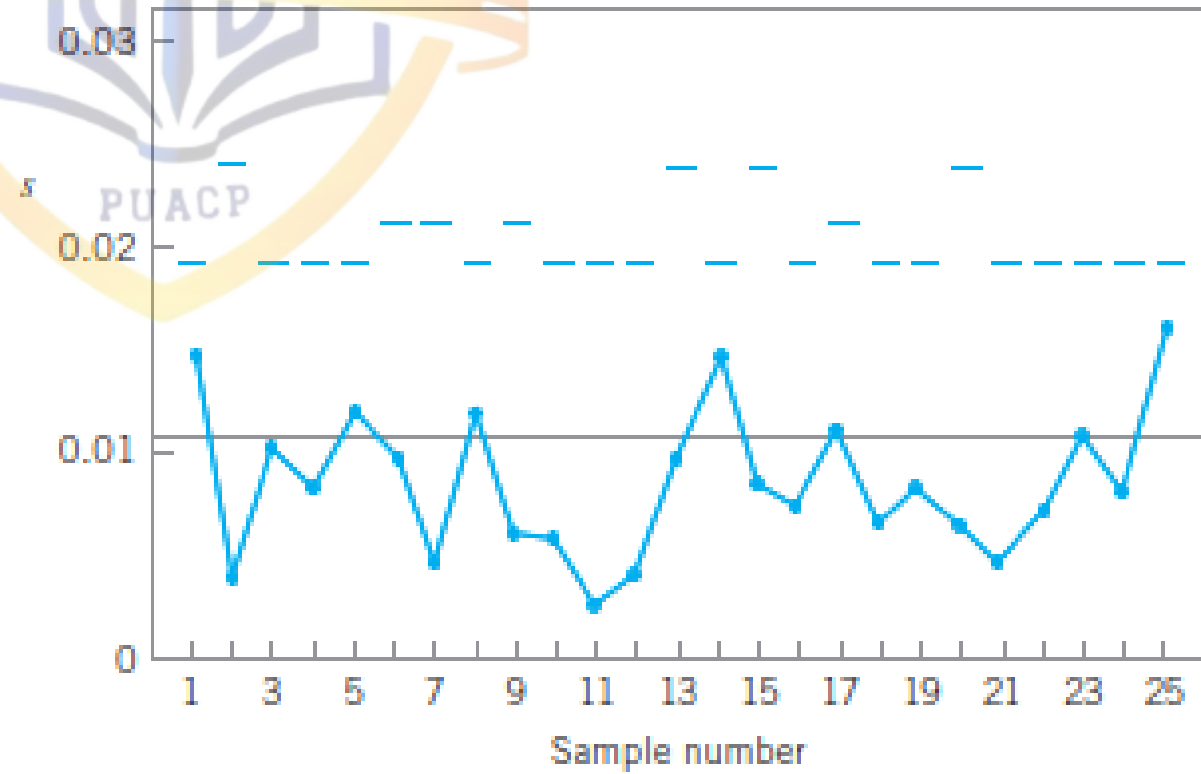
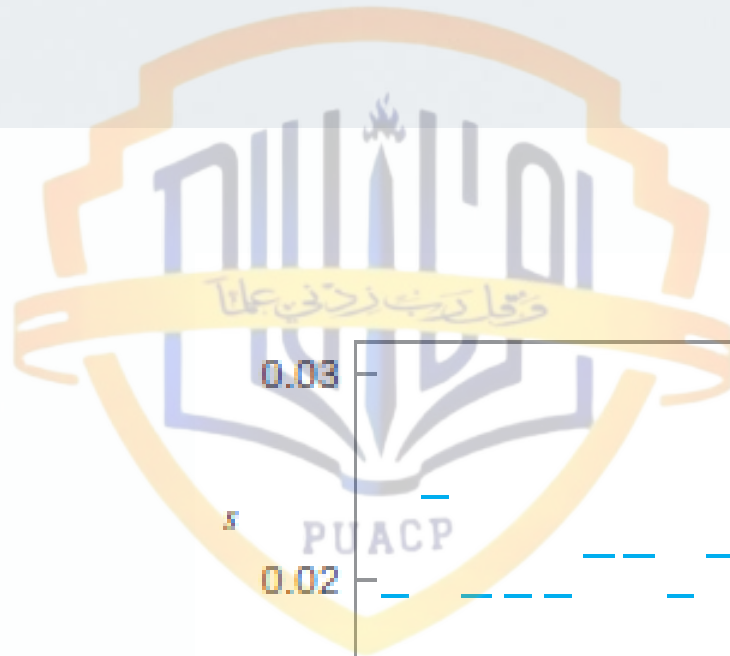
$$CL = 0.0103$$

$$LCL = 0(0.0103) = 0$$

Computation of Control Limits for $\bar{x}$ and s Charts with Variable Sample Size

Sample	n	$\bar{x}$	s	A_3	$\bar{x}$ Chart		B_3	B_4	s Chart	
					LCL	UCL			LCL	UCL
1	5	74.010	0.0148	1.427	73.986	74.016	0	2.089	0	0.022
2	3	73.996	0.0046	1.954	73.981	74.021	0	2.568	0	0.026
3	5	74.008	0.0147	1.427	73.986	74.016	0	2.089	0	0.022
4	5	74.003	0.0091	1.427	73.986	74.016	0	2.089	0	0.022
5	5	74.003	0.0122	1.427	73.986	74.016	0	2.089	0	0.022
6	4	73.996	0.0099	1.628	73.984	74.018	0	2.266	0	0.023
7	4	73.999	0.0055	1.628	73.984	74.018	0	2.266	0	0.023
8	5	73.997	0.0123	1.427	73.986	74.016	0	2.089	0	0.022
9	4	74.004	0.0064	1.628	73.984	74.018	0	2.266	0	0.023
10	5	73.998	0.0063	1.427	73.986	74.016	0	2.089	0	0.022
11	5	73.994	0.0029	1.427	73.986	74.016	0	2.089	0	0.022
12	5	74.001	0.0042	1.427	73.986	74.016	0	2.089	0	0.022
13	3	73.994	0.0100	1.954	73.981	74.021	0	2.568	0	0.026
14	5	73.990	0.0153	1.427	73.986	74.016	0	2.089	0	0.022
15	3	74.008	0.0087	1.954	73.981	74.021	0	2.568	0	0.026
16	5	73.997	0.0078	1.427	73.986	74.016	0	2.089	0	0.022
17	4	73.999	0.0115	1.628	73.984	74.018	0	2.226	0	0.023
18	5	74.007	0.0070	1.427	73.986	74.016	0	2.089	0	0.022
19	5	73.998	0.0085	1.427	73.986	74.016	0	2.089	0	0.022
20	3	74.008	0.0068	1.954	73.981	74.021	0	2.568	0	0.026
21	5	74.000	0.0122	1.427	73.986	74.016	0	2.089	0	0.022
22	5	74.002	0.0074	1.427	73.986	74.016	0	2.089	0	0.022
23	5	74.002	0.0119	1.427	73.986	74.016	0	2.089	0	0.022
24	5	74.005	0.0087	1.427	73.986	74.016	0	2.089	0	0.022
25	5	73.998	0.0162	1.427	73.986	74.016	0	2.089	0	0.022







Thank You